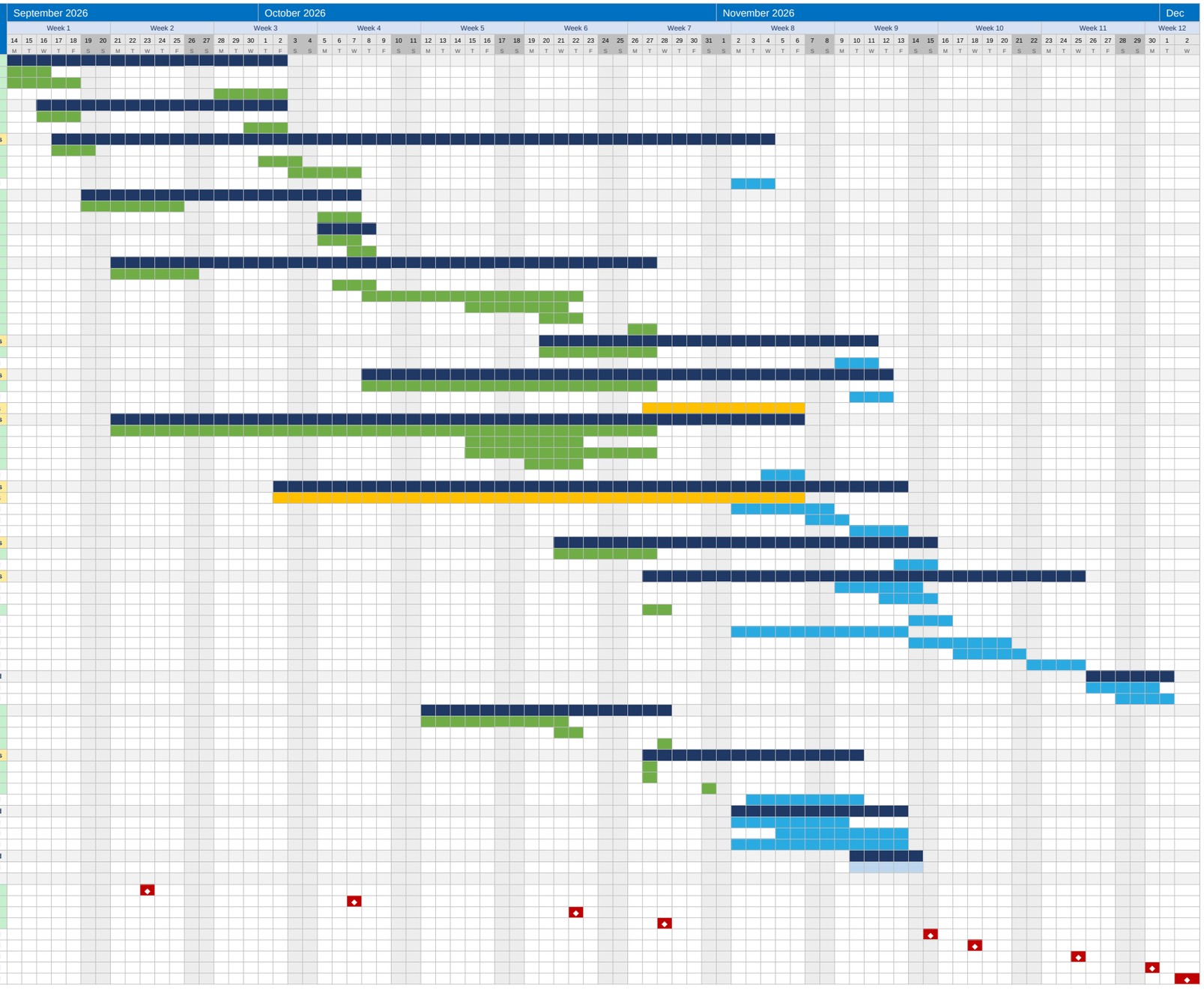
FitForm AI - Project Gantt Chart (daily), IMP Part B

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Task Name	Outline week	Start	End	Days	Status
1. Set up Python 3.11 virtual environment (Raspberry Pi OS + laptop)	W1	14 Sep	2 Oct	19	Done
Install Raspberry Pi OS on the microSD card; first boot of the Pi 5		14 Sep	16 Sep	3	Done
Laptop Python environment; fix library version error		14 Sep	18 Sep	5	Done
Pi 5 Python 3.12 virtual environment with uv (OS ships Python 3.13)		28 Sep	2 Oct	5	Done
2. Install and configure OpenCV	W2	16 Sep	2 Oct	17	Done
Install OpenCV 4.11 on the laptop; test the webcam		16 Sep	18 Sep	3	Done
Install OpenCV on the Pi 5; remove conflicting builds		30 Sep	2 Oct	3	Done
3. Install and configure MediaPipe; test camera feed and GPIO wiring for LED/buzzer	W3	17 Sep	4 Nov	49	In progress
Install MediaPipe on the laptop; test pose on the webcam		17 Sep	19 Sep	3	Done
Install MediaPipe 0.10.18 on the Pi 5; run the logic tests		1 Oct	3 Oct	3	Done
Pi camera support (rpicam) and AI Camera firmware		3 Oct	7 Oct	5	Done
GPIO wiring test of the LEDs and buzzer on the breadboard (moved)		2 Nov	4 Nov	3	Not started
4. Develop MediaPipe Pose landmark extraction (shoulder, elbow, wrist, hip coordinates)	W4	19 Sep	7 Oct	19	Done
Restructure the prototype into modules (pose_engine.py)		19 Sep	25 Sep	7	Done
Use MediaPipe 3D world landmarks (view-independent)		5 Oct	7 Oct	3	Done
5. Implement joint-angle calculation functions (elbow, torso, wrist angles)	W5	5 Oct	8 Oct	4	Done
Elbow angle and upper-arm drift check (replaces hip swing check)		5 Oct	7 Oct	3	Done
Wrist angle from the hand knuckles (MediaPipe Hands)		7 Oct	8 Oct	2	Done
6. Build the bicep curl rep-counting state machine with form-validation checks	W6	21 Sep	27 Oct	37	Done
State machine for left and right arms (laptop test)		21 Sep	26 Sep	6	Done
Clenched-fist check with MediaPipe Hands		6 Oct	8 Oct	3	Done
Fix live rep counting on the Pi using the debug view		8 Oct	22 Oct	15	Done
Fist direction: per-arm calibration from labelled runs; tip instead of reject		15 Oct	21 Oct	7	Done
Pause after a rejected rep with fix text; walk-away ends the session		20 Oct	22 Oct	3	Done
Wrist-height tempo and lowering-speed check		26 Oct	27 Oct	2	Done
7. Test OpenCV/MediaPipe detection under varying lighting conditions	W7	20 Oct	11 Nov	23	In progress
Record test clips for repeatable testing (tools/record_clips.py)		20 Oct	27 Oct	8	Done
Lighting tests (bright, dim, backlit) and record results		9 Nov	11 Nov	3	Not started
8. Test detection under varying distances/backgrounds; tune thresholds for stability	W8	8 Oct	12 Nov	36	In progress
Tune angle, visibility and confidence thresholds		8 Oct	27 Oct	20	Done
Distance and background tests and record results		10 Nov	12 Nov	3	Not started
Frame-rate check on the Pi 5 (CNN every 3rd frame, camera watchdog)		27 Oct	6 Nov	11	In progress
9. Develop supporting modules: display overlay, GPIO feedback, calorie tracking, session logging (CSV)	W9	21 Sep	6 Nov	47	In progress
On-screen overlay, debug view, start-up demo and full screen		21 Sep	27 Oct	37	Done
Calorie estimate (MET 3.5, 2024 Compendium) and session summary		15 Oct	22 Oct	8	Done
CSV session log and Flask phone app: accounts, QR login, history		15 Oct	27 Oct	13	Done
Goal for the next session from guidelines (ACSM 2009, WHO 2020)		19 Oct	22 Oct	4	Done
GPIO feedback module with the LEDs and buzzer (moved)		4 Nov	6 Nov	3	Not started
10. Design and 3D print the Raspberry Pi/display/LED enclosure (Bambu Lab X1 Carbon, PLA)	W10	2 Oct	13 Nov	43	In progress
Enclosure CAD in Fusion 360 (tray, port openings, hinged lid)		2 Oct	6 Nov	36	In progress
Polished design (supervisor): rounded shape, camera mounts, cable routing, logo		2 Nov	8 Nov	7	Not started
Test-print one section and check the fit		7 Nov	9 Nov	3	Not started
Print the full enclosure and finish it (sanding, two colours)		10 Nov	13 Nov	4	Not started
11. Assemble hardware into the enclosure; integrate pipeline on Raspberry Pi 5 (camera, GPIO, Flask endpoint)	W11	21 Oct	15 Nov	26	In progress
Integrate the software on the Pi (both cameras, Flask app)		21 Oct	27 Oct	7	Done
Mount the Pi, cameras, LEDs and buzzer in the enclosure		13 Nov	15 Nov	3	Not started
12. End-to-end system testing and accuracy/reliability evaluation; write and finalise report; submission buffer	W12	27 Oct	25 Nov	30	In progress
Repeated test sessions: automated vs manually counted reps		9 Nov	14 Nov	6	Not started
Fix issues found; software and hardware complete (code freeze)		12 Nov	15 Nov	4	Not started
Demo videos on Google Drive (bicep curl, exit session)		27 Oct	28 Oct	2	Done
Final demo video and results for the report		14 Nov	16 Nov	3	Not started
Report: introduction, literature review, methodology		2 Nov	13 Nov	12	Not started
Report: results, discussion, conclusions		14 Nov	20 Nov	7	Not started
Report: project management and quality management review		17 Nov	21 Nov	5	Not started
Proofread, similarity check (<20%), submit (buffer)		22 Nov	25 Nov	4	Not started
13. Prepare VIVA slides and rehearse the demo (added)	-	26 Nov	1 Dec	6	Not started
Prepare the 8 VIVA slides		26 Nov	30 Nov	5	Not started
Rehearse the presentation and live demo		28 Nov	1 Dec	4	Not started
14. Two-camera approach (added)	-	12 Oct	28 Oct	17	Done
AI Camera at the front + USB webcam at the side; fist/palm from both views		12 Oct	21 Oct	10	Done
Find the cameras automatically (front / side)		21 Oct	22 Oct	2	Done
Review at Session 3: fist direction left as a limitation		28 Oct	28 Oct	1	Done
15. Weight-in-hand CNN (added)	-	27 Oct	10 Nov	15	In progress
Collect labelled hand crops (about 9,500, front + side)		27 Oct	27 Oct	1	Done
Fine-tune MobileNetV3-Small, export ONNX, run with OpenCV (98.5 %)		27 Oct	27 Oct	1	Done
Feasibility test: AI Camera on-chip object detector		31 Oct	31 Oct	1	Done
Round 3 data: 3D-printed dumbbell, other objects and people		3 Nov	10 Nov	8	Not started
16. Extra features suggested by the supervisor at Session 3 (added)	-	2 Nov	13 Nov	12	Not started
Personalised session report (PDF from the phone app)		2 Nov	9 Nov	8	Not started
Speech-to-text voice commands (start, stop, summary) - offline		5 Nov	13 Nov	9	Not started
Polished 3D chassis - see task 10		2 Nov	13 Nov	12	Not started
17. Battery power - if time allows (added)	-	10 Nov	14 Nov	5	Not started
Battery path (fuse, SB560 diode, HW-688) and load test (optional)		10 Nov	14 Nov	5	Not started
Milestones					
◆ **Session 1 - Project review 1 (Logbook 5)**		23 Sep	23 Sep	0	Done
◆ **Session 2 - Project review 2 (Logbook 6)**		7 Oct	7 Oct	0	Done
◆ **Software target - reps counted reliably on the Pi**		22 Oct	22 Oct	0	Done
◆ **Session 3 - Project review 3 (Logbook 7)**		28 Oct	28 Oct	0	Done
◆ **Software and hardware complete (code freeze)**		15 Nov	15 Nov	0	Not started
◆ **Session 4 - Project review 4 (Logbook 8)**		18 Nov	18 Nov	0	Not started
◆ **Final project report submission**		25 Nov	25 Nov	0	Not started
◆ **VIVA slides submission**		30 Nov	30 Nov	0	Not started
◆ **Project defence (VIVA) and demo**		2 Dec	2 Dec	0	Not started



Legend

- Heading (outline task) - span of its subtasks
- Done
- In progress
- Planned
- Optional / still in development
- Milestone
- Weekend

Comparison with the Project Outline Gantt chart (Appendix A)

Term week 1 = 14-20 Sep 2026, so outline week Wn = the n-th week from 14 Sep.

Outline task	Outline week (dates)
Set up Python 3.11 virtual environment (Raspberry Pi OS + laptop)	W1 (14 Sep - 20 Sep)
Install and configure OpenCV	W2 (21 Sep - 27 Sep)
Install and configure MediaPipe; test camera feed and GPIO wiring for LED/buzzer	W3 (28 Sep - 04 Oct)
Develop MediaPipe Pose landmark extraction (shoulder, elbow, wrist, hip coordinates)	W4 (05 Oct - 11 Oct)
Implement joint-angle calculation functions (elbow, torso, wrist angles)	W5 (12 Oct - 18 Oct)
Build the bicep curl rep-counting state machine with form-validation checks	W6 (19 Oct - 25 Oct)
Test OpenCV/MediaPipe detection under varying lighting conditions	W7 (26 Oct - 01 Nov)
Test detection under varying distances/backgrounds; tune thresholds for stability	W8 (02 Nov - 08 Nov)
Develop supporting modules: display overlay, GPIO feedback, calorie tracking, session logging (CSV)	W9 (09 Nov - 15 Nov)
Design and 3D print the Raspberry Pi/display/LED enclosure (Bambu Lab X1 Carbon, PLA)	W10 (16 Nov - 22 Nov)
Assemble hardware into the enclosure; integrate pipeline on Raspberry Pi 5 (camera, GPIO, Flask endpoint)	W11 (23 Nov - 29 Nov)
End-to-end system testing and accuracy/reliability evaluation; write and finalise report; submission buffer	W12 (30 Nov - 06 Dec)
Two-camera approach (added)	-
Weight-in-hand CNN (added)	-
Supervisor's extra features (added)	-
VIVA slides and rehearsal (added)	-

Updated span	Status (31 Oct)
14 Sep - 02 Oct	Done
16 Sep - 02 Oct	Done
17 Sep - 04 Nov	In progress
19 Sep - 07 Oct	Done
05 Oct - 08 Oct	Done
21 Sep - 27 Oct	Done
20 Oct - 11 Nov	In progress
08 Oct - 12 Nov	In progress
21 Sep - 06 Nov	In progress
02 Oct - 13 Nov	In progress
21 Oct - 15 Nov	In progress
27 Oct - 25 Nov	In progress
12 Oct - 28 Oct	Done
27 Oct - 10 Nov	In progress
02 Nov - 13 Nov	Not started
26 Nov - 01 Dec	Not started

Change and reason
Took longer: Raspberry Pi OS Trixie ships Python 3.13, which MediaPipe does not support, so a Python 3.12 environment was made with uv.
Laptop done in W1; the Pi part was done together with task 1 in W3.
MediaPipe and camera done (AI Camera needed its firmware). GPIO wiring moved to early November to keep the focus on the software (critical path).
Done early on the laptop (W1-W2); 3D landmarks added in W4.
Done in W4; wrist angle from the hand knuckles; hip angle replaced by an upper-arm drift check because the hips are often out of view.
Started early on the laptop (W2); live counting on the Pi took until 22 Oct. Fist direction could not be judged reliably from one camera; agreed with the supervisor to leave it as a limitation (advice only).
Test clips recorded in W6 so every change is tested on the same video; the formal lighting tests are run in W9 with the finished system.
Threshold tuning started early (W4) because counting did not work on the Pi; distance/background tests in W9.
Grew in scope: the CSV log became a Flask phone app with accounts, QR login, session history, calorie chart and a goal for the next session (W5-W7). GPIO feedback moved to W8.
CAD started early but paused while the software was fixed; the supervisor asked for a polished chassis, so the redesign and printing are W8-W9.
Software integration on the Pi finished in W6-W7 (both cameras, phone app); mounting in the enclosure in W9.
Demo videos already recorded (W7). Report writing starts in W8 so it is not left to the end; code freeze 15 Nov, report due 25 Nov.
Added to judge the fist and palm from the side; reviewed at Session 3.
Added to check that a weight is held; 98.5 % on a new session; more data with a dumbbell next.
Personalised report, speech-to-text and a polished chassis, suggested at Session 3.
Not in the outline; slides due 30 Nov, VIVA 2 Dec.